

Agniveer Vayu Science (Group X) - 19 Jan 2023 - Memory Based Paper

Physics

Question 1

If 'F' is the force acting on a particle of mass 'm' performing simple harmonic motion, then the time period of the simple harmonic motion is equal to _____.

Options:

- A. $2\pi \sqrt{(k/m)}$
- B. $2\pi \sqrt{(m/k)}$
- C. $1/(2\pi) \times \sqrt{(m/k)}$
- D. $1/(2\pi) \times \sqrt{(k/m)}$

Answer: B

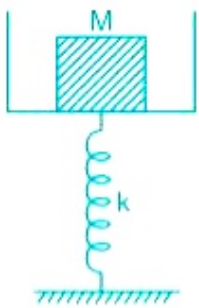
Solution:

CONCEPT:

- **Simple Harmonic Motion (SHM):** Simple harmonic motion is a special type of periodic motion or oscillation where the restoring force is directly proportional to the displacement and acts in the direction opposite to that of displacement.
 - **Example:** Motion of an undamped pendulum, undamped spring-mass system.
- **Time period (T):** The time taken by the particle to complete one revolution is called time period.

The time period (T) of a spring block system doing the simple harmonic motion is given by:

Here force (F) is equal to the weight (mg) of the block.



$$ma = -k x$$

$$a = -(k/m) x$$

$$T = 2 \pi \sqrt{\frac{m}{k}}$$

Where m is mass and k is spring constant.

EXPLANATION:

The time period of the simple harmonic motion is given by

$$T = 2\pi \sqrt{(m/k)}$$

So option 2 is correct.

Question 2

Velocity of sound is maximum in:

Options:

A. hydrogen

B. vacuum

C. steel

D. water

Answer: C

Solution:

Concept:

- The speed of the sound depends on the **density of the medium** through which it is travelling.
- The particles in the **solid medium** are closely packed. Hence, the density is high. Information from one molecule to another molecule travels faster in the solid medium.
- **In liquid**, the molecules are farther apart than in solids. Hence, the density is low for liquids.

- In gas, the molecules are very loosely dispersed. The speed of the sound wave will be maximum in steel.

Explanation:

- Sound is a **mechanical wave** that needs a medium to travel.
- The speed of sound is **maximum in solid**, then it is in liquid and it is the slowest in gases.
- The speed of sound remains almost the same for all frequencies in a given medium under the same physical conditions.

From the experiments these data are obtained:

- The velocity of sound waves in **Hydrogen is 1270 m/s**.
- The velocity of sound waves in **steel is 5920 m/s**.
- The velocity of sound waves in a **vacuum is zero**.
- The velocity of sound waves in **water is 1480 m/s**.

So, velocity of sound is maximum in **steel** among them.

Question 3

The dimension of Planck's constant is ____.

Options:

- A. [MLT]
- B. $[ML^1T^{-1}]$
- C. $[ML^2T^{-1}]$
- D. $[ML^2T^{-2}]$

Answer: C

Solution:

Concept:

- Measurement of any physical quantity involves comparison with a certain basic, arbitrarily chosen, internationally accepted reference standard called **unit and Dimension** is a mathematical tool used for studying the nature of physical quantities.
- The basic concept of dimensions is that we can add or subtract only those quantities which have the same dimensions.
- And the **dimensional formula** is defined as the expression of the physical quantity in terms of mass, length, and time.
- Energy of electromagnetic waves is given by dimension analysis and it is represented by formula $E = hf$

Explanation:

We know that the energy of an emitted photon is given as $E = hf$

Where E is the energy of photon and h is plank's constant and ν is the frequency of an emitted photon

$$\therefore h = E/\nu$$

Hence the dimension of Plank's constant is given as,

$$h = \frac{E}{\nu} = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

$$\Rightarrow \text{Dimension of } h = [ML^2T^{-1}]$$

Question 4

A magnet of magnetic moment M is situated with its axis along the direction of a magnetic field of strength B . The work done in rotating it by an angle of 180° will be:

Options:

- A. -MB
- B. +MB
- C. +2MB
- D. Zero

Answer: C

Solution:

CONCEPT:

Magnetic dipole:

- A Magnetic Dipole consists of **two unlike poles of equivalent strength and separated by a small distance**.

Magnetic moment:

- It is defined as the **product of pole strength and the distance between the poles of a magnet**.

$$\Rightarrow M = m \times 2l$$

where M = magnetic moment, m = pole strength and $2l$ = distance between the poles

- Work done in rotating a magnet of magnetic moment M with angle θ in an external magnetic field B is given as,

$$\Rightarrow W = MB (1 - \cos\theta)$$

CALCULATION:

Given $\theta = 180^\circ$

- The work done in rotating a magnet of magnetic moment M by an angle of 180° in the magnetic field B is given as,

$$\Rightarrow W = MB (1 - \cos\theta)$$

$$\Rightarrow W = MB (1 - \cos 180)$$

$$\Rightarrow W = MB (1 + 1)$$

$$\Rightarrow W = 2MB$$

- Hence, option 3 is correct.
-

Question 5

Two coplanar concurrent forces of magnitude 3 kN and 4 kN makes an angle 60° with each other, then the magnitude of resultant is:

Options:

A. 4.03 kN

B. 3.25 kN

C. 6.08 kN

D. 7 kN

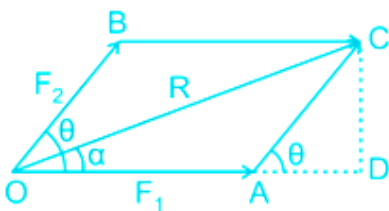
Answer: C

Solution:

Law of Parallelogram of forces:

This law is used to determine the **resultant of two coplanar forces acting at a point.**

It states that “If two forces acting at a point are represented in magnitude and direction by two adjacent sides of a parallelogram, then their resultant is represented in magnitude and direction by the diagonal of the parallelogram which passes through that common point.”



Let two forces F_1 and F_2 , acting at the point O be represented, in magnitude and direction, by the directed line OA and OB inclined at an angle θ with each other.

Then if the parallelogram OACB be completed, the resultant force R will be represented by the diagonal OC.

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

Calculation:

$$F_1 = 3 \text{ kN}, F_2 = 4 \text{ kN}, \theta = 60^\circ$$

$$F_R = (3^2 + 4^2 + 2 \times 3 \times 4 \cos 60^\circ)^{1/2}$$

$$F_R = 6.08 \text{ kN}$$

Question 6

When a beam of white light incident on a prism undergoes dispersion then the _____ of the constituent colors of white light.

Options:

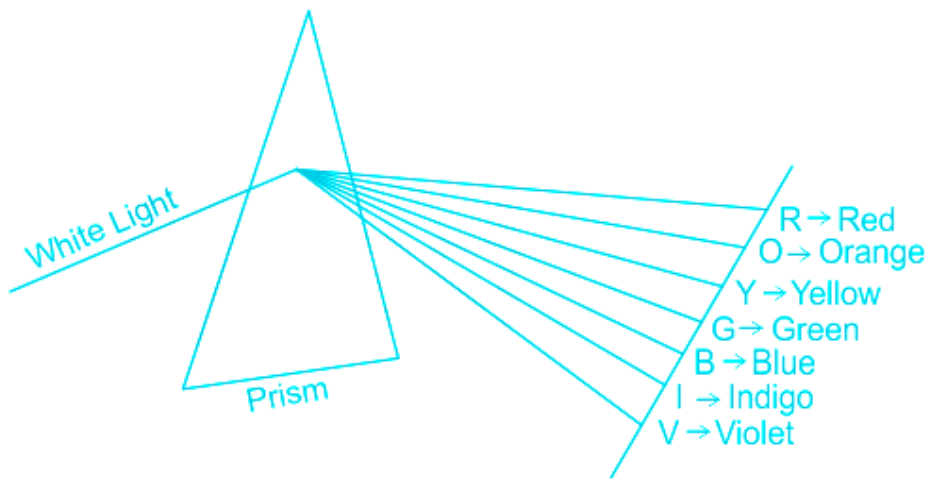
- A. red colour bends the most as it has the lowest wavelength
- B. violet colour bends the most as it has the lowest wavelength
- C. violet colour bends the most as it has the highest wavelength
- D. violet colour bends the most as it has the lowest frequency

Answer: B

Solution:

CONCEPT:

- **Dispersion of light:** The splitting of a composite beam of light into its constituent colours is called **dispersion of light**.



- The dispersion of white light occurs because the colours of white light travel at different speeds through the glass prism.
 - The dispersion was discovered by Sir Issac Newton.
 - Newton discovered that light is made up of seven different colours.

EXPLANATION:

- From the above discussion, we can say that When a beam of white light incident on a prism undergoes dispersion then the violet colour bends the most as it has the lowest wavelength of the constituent colors of white light. So option 2 is correct.

Question 7

Which of the following is correct regarding Faraday's Law?

Options:

A.

Change in magnetic flux can cause a voltage to be induced.

B.

The induced emf can not generate the current.

C.

Increase in number of turns in a coil will decrease induced emf

D.

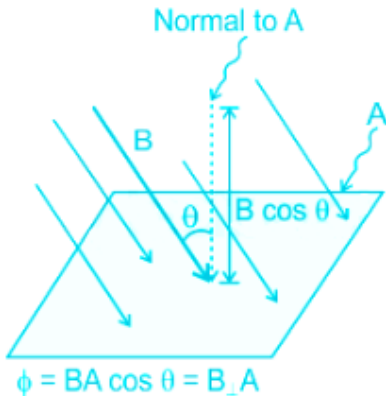
All of the above statements are correct.

Answer: A

Solution:

CONCEPT:

- **Magnetic flux** is the total magnetic field that passes through the given area.
- If we choose a simple flat surface with area A and there is an angle θ between the normal to the surface and a magnetic field vector (magnitude B) then the **magnetic flux** is $\phi = BA \cos \theta$



- **Faraday's Law:** Any change in the magnetic flux of wire will cause a voltage (**induced emf**) to be "induced" in the coil.
- This change can be produced by changing the magnetic field strength, moving the coil into or out of the magnetic field, moving a magnet toward or away from the coil, rotating the coil relative to the magnet, etc.

$$\text{Induced emf} = -N \frac{\Delta \phi}{\Delta t}$$

CALCULATION:

- **Option 1:** Change in magnetic flux $\Delta \phi$ can induce emf as there is term $\Delta \phi$ in the formula of induced emf.
 - **So option 1 is a correct statement.**
- **Option 2:** This induced emf works as a potential difference and can generate current in a wire.
 - **So option 2 is an incorrect statement.**
- **Option 3:** The magnitude of the Induced emf is calculated by

$$\text{Induced emf} = N \frac{\Delta \phi}{\Delta t}$$

- So increase in the number of turns N will increase the magnitude of the induced emf.
 - **So option 3 is an incorrect statement.**
 - So the correct answer is **option 1**.
-

Question 8

The Carnot's engine working between 400 K and 800 K has a work output of 1200 J per cycle. The amount of heat energy supplied to the engine from the source in each cycle is:

Options:

A. 2400 J

B. 1600 J

C. 3200 J

D. 1800 J

Answer: A

Solution:

CONCEPT:

→The efficiency of the heat engine is written as;

$$\eta = 1 - \frac{T_2}{T_1}$$

Here we have T_2 and T_1 are temperatures.

→The efficiency of the heat engine is the ratio of the work done to the heat and it is written as;

$$\eta = \frac{W}{Q}$$

Here we have W as the work done and Q as the heat.

CALCULATION:

Given: Temperature, $T_1 = 800$ K, $T_2 = 400$ K, Work, $W = 1200$ J

As we know,

$$\eta = 1 - \frac{400}{800}$$

$$\Rightarrow \eta = \frac{400}{800}$$

$$\Rightarrow \eta = \frac{1}{2}$$

$$\text{Now, } \eta = \frac{W}{Q}$$

On putting all the values we have;

$$\frac{1}{2} = \frac{1200}{Q}$$

$$\Rightarrow Q = 2400 \text{ J}$$

Hence option 1) is the correct answer.

Question 9

if kinetic energy of a 2 kg mass object has been changed by 3 kJ by a 1000 N force acting along the line of motion on a smooth surface, then

displacement made by the object is _____

Options:

A. 2 m

B. 3 m

C. 1 m

D. 4 m

Answer: B

Solution:

Concept:

Work

- **Work is said to be done** by a force when the **body is displaced actually through some distance in the direction of the applied force.**
- Since the **body is being displaced in the direction of F**, therefore **work done by the force in displacing the body through a distance s** is given by

$$W = \vec{F} \cdot \vec{s}$$

$$\text{Or, } W = Fs \cos \theta$$

Work-Energy Theorem:

- The **work-energy theorem** states that the **net work** done by the **forces** on an object is **equal to** the change in its **kinetic energy**.

$$\Rightarrow W = \Delta KE$$

Where W = work done and ΔKE = change in kinetic energy

Calculation:

Given,

$$\text{Change in Kinetic energy} = \Delta KE = 3 \text{ kJ} = 3000 \text{ J}$$

$$\text{Force is along the line of motion } F = 1000 \text{ N}$$

$$\text{The angle between the force and displacement } \theta = 0^\circ$$

By work energy theorem

$$W = Fs \cos \theta = 3000 \text{ J}$$

$$1000 \text{ N} \times s \cos 0^\circ = 3000 \text{ J}$$

$$1000 \text{ N} \times s = 3000 \text{ J}$$

$$s = 3 \text{ m}$$

So displacement is 3 m.

Question 10

One feels heavier in a lift when the lift

Options:

- A. is going down steadily
- B. just begin to go up
- C. is moving up steadily
- D. decends freely

Answer: B

Solution:

Concept-

Apparent weight of a person-

(i) When the lift moves upward with acceleration a . Then the net upward force on the person is-

$$R - mg = ma$$

$$\therefore \text{Apparent weight, } R = mg + ma = m(g + a)$$

So when a lift accelerates upwards, the apparent weight of the person inside it increases.

(ii) When the lift moves downwards with acceleration a . Then the net downward force on the person is-

$$Mg - R = ma$$

$$\therefore \text{Apparent weight, } R = mg - ma = m(g - a)$$

So when a lift accelerates downwards, the apparent weight of the person inside it decreases.

(iii) When the lift is at rest or moving with uniform velocity v downward/upward. The acceleration $a = 0$, then the net force on the person is-

$$R - mg = m \times 0 = 0$$

$$R = mg$$

$$\therefore \text{Apparent weight} = \text{Actual weight}$$

(iv) When the lift falls freely. If the supporting cable of the lift breaks, the lift falls freely under gravity. Then $a = g$. The net downward force on the person is -

$$R = m(g - g) = 0.$$

Thus the apparent weight of the man becomes zero. This is because both the man and the lift are moving downwards with same acceleration 'g' and so there are no forces of action and reaction between the person and the lift. Hence a person develops a feeling of **weightlessness** when he falls freely under gravity.

Explanation -

When the lift moves upward with acceleration a -

Then the net upward force on the person is-

$$R - mg = ma$$

$$\therefore \text{Apparent weight, } R = mg + ma = m(g + a)$$

One feels heavier in a lift when the lift just begins to go up because our body gains inertia from the position of rest and pushes up against the gravity so here the weight becomes zero and our mass makes us feel heavier. Thus, so when a lift accelerates upwards, the apparent weight of the person inside it increases.

Question 11

The correct order of electromagnetic spectrum with decreasing frequency is:

Options:

- A. Microwaves, Radiowaves, Infrared rays, Ultraviolet rays, X-rays
- B. Radiowaves, Infrared rays, Ultraviolet rays, Microwaves, X-rays
- C. X-rays, Infrared rays, Microwaves, Radiowaves, Ultraviolet rays
- D. X-rays, Ultraviolet rays, Infrared rays, Microwaves, Radiowaves

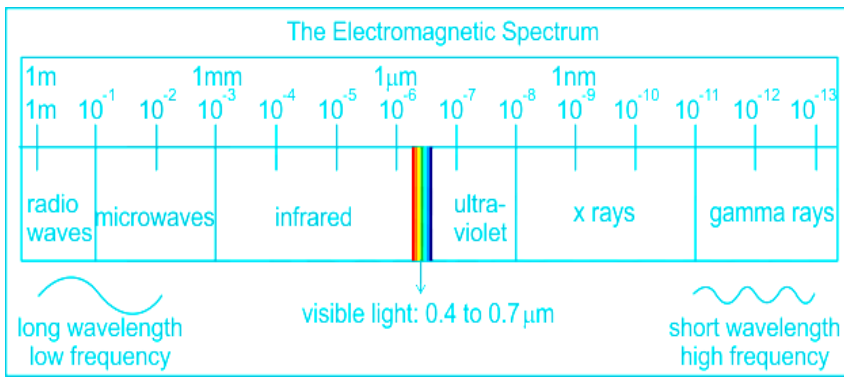
Answer: D

Solution:

Electromagnetic (EM)spectrum:

The electromagnetic (EM)spectrum is the range of all types of EM radiation arranged according to frequency or wavelength.

- In order of decreasing wavelength or increasing frequency these are:
- radiowaves radiation, microwave radiation, infrared, visible light, ultraviolet, X-rays, and gamma rays.
-



Analysis

The correct order of electromagnetic spectrum with decreasing frequency is:

X-rays > Ultraviolet rays > Infrared rays > Microwaves > Radiowaves

Question 12

What is the unit of the pole strength of a magnet?

Options:

- A. A-m
- B. A-m²
- C. A/m²
- D. A/m

Answer: A

Solution:

CONCEPT:

- **Pole strength:** When another magnet of opposite polarity is brought near a pole, the amount of force a pole of magnet exerts is called pole strength.
 - In SI units, there are two possible units for magnetic pole strength
 1. It can be defined as the force per unit H field, in which case the SI unit is the **weber (Wb)**.
 2. If you define it as force per unit B field, then the SI unit is the **ampere-meter (A m)**.

EXPLANATION:

- The **unit of pole strength is Ampere-meter (A-m)** and weber is the SI unit system.
- So the correct answer is **option 1**.

Question 13

The wavelength of a light in a medium is 400 nm and the frequency of the light is 3×10^{14} Hz. Find the refractive index of light in that medium.

Options:

A. 1.5

B. 2.5

C. 1.33

D. 2

Answer: B

Solution:

Concept:

- The distance between one crest and the other crest or the distance between one troughs to another is called as **wavelength of the wave**.
- **Wavelength** is denoted by λ .
- The number of waves which passes through a fixed point in a given time is called as **frequency** of the wave. It is denoted by f .

Refractive index:

- The ratio of speed of light in vacuum to speed of light in a medium is called as **the refractive index** of that medium. It is also called as **absolute refractive index**.

$$\text{Refractive index } (\mu) = \frac{\text{Speed of light in vacuum}(C)}{\text{Speed of light in a medium}(v)}$$

$$\text{Refractive index } (\mu) = \frac{C}{f \times \lambda}$$

Where f is frequency, λ is wavelength and C is speed of light $= 3 \times 10^8$ m/s

Explanation:

Given that;

$$\text{Frequency} = f = 3 \times 10^{14} \text{ Hz}$$

$$\text{Wavelength} = \lambda = 400 \text{ nm} = 400 \times 10^{-9} \text{ m}$$

$$C = 3 \times 10^8 \text{ m/s}$$

$$\mu = \frac{3 \times 10^8}{3 \times 10^{14} \times 400 \times 10^{-9}} = \frac{10^8}{400 \times 10^5} = 2.5$$

$$\text{Refractive index} = \mu = 2.5$$

Question 14

Till what temperature (in °C) should a gas at 27°C be heated so as to increase its volume by 40% keeping pressure constant?

Options:

A. 400°C

B. 147°C

C. 200°C

D. 247°C

Answer: B

Solution:

CONCEPT:

- **Ideal gas:** It is a hypothetical gas, consisting of molecules that are having **negligible volume** and their **collision is perfectly elastically** (means no loss of energy) and **obeys gas law**.

The **Equation of State for an ideal gas** is given by:

$$\Rightarrow PV = \mu RT$$

Where

μ = the number of moles of gas,

P = Pressure

T = Temperature

V = Volume

R = the gas constant.

Calculation:

Given that,

$$P_1 = P_2$$

$$V_2 = \text{increases by 40\% of } V_1$$

$$\text{i.e., } V_2 = 1.4V_1$$

And from the ideal gas equation, we can see that

$$P_1 V_1 = \mu R T_1 \text{ \& } P_2 V_2 = \mu R T_2$$

$$\therefore P_1 = P_2 \Rightarrow \frac{\mu R T_1}{V_1} = \frac{\mu R T_2}{V_2}$$

$$\frac{T_1}{V_1} = \frac{T_2}{V_2} \Rightarrow \frac{27 + 273}{V_1} = \frac{T_2}{1.4 \times V_1}$$

$$T_2 = 1.4 \times (27 + 273) = 420 - 273 = 147^\circ \text{ C}$$

Extra points:

The properties of an ideal gas are:

- They are having **negligible volume, point size molecules**.
- There is **no attractive or repulsive force** between the particles of gas
- The **collision** between the gases and the walls of the container should be **perfectly elastic** (there is no loss of energy).

The Equation of State is an equation that connects all the state variables of a system.

Question 15

Consider a circular ring of mass 1 kg and diameter 0.2 m. It is making 10 rotations per second about an axis passing through its centre and normal to the surface. The value of angular momentum is:-

Options:

A.

$$0.628 \text{ kg m}^2/\text{sec}$$

B.

$$0.4 \text{ kg m}^2/\text{sec}$$

C.

$$1.256 \text{ kg m}^2/\text{sec}$$

D.

$$0.2 \text{ kg m}^2/\text{sec}$$

Answer: A

Solution:

Concept:

- Angular momentum (**L**) or rotational momentum is the **moment of inertia times (I)angular velocity (ω)**.
- Angular momentum is the Vector quantity and its directly proportional to the torque applied to the body.
- When the direction of torque and direction of rotation is the same then angular momentum increases and when the direction of torque and rotation is opposite then angular momentum decreases.

$$\therefore \mathbf{L} = \mathbf{I} \times \boldsymbol{\omega}$$

Where,

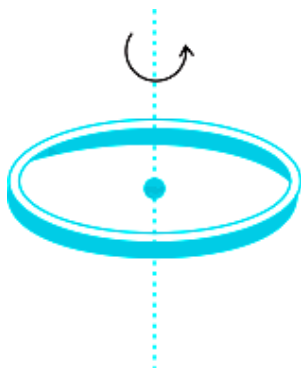
L is angular momentum
and, ω is angular velocity

Explanation:

Given data,

Circular ring of mass (**m**) = 1 kg,

diameter (d) = 0.2 m $\rightarrow$ radius (**r**) = 0.1 m



- As we know the angular momentum of a ring for which axes of rotation pass from its center and normal to the plane is **$\mathbf{I} = \mathbf{mr}^2$**
- As the ring rotates 10 revolutions per second it means it has angular velocity (**ω**) = $10 \times 2\pi \frac{\text{rad}}{\text{s}}$

Now Angular momentum (**L**) = **$\mathbf{I} \times \boldsymbol{\omega}$**

$$\therefore \mathbf{L} = \mathbf{mr}^2 \times \boldsymbol{\omega}$$

$$\therefore \mathbf{L} = (1) \cdot (0.1)^2 \times 20\pi$$

$$\therefore \mathbf{L} = 0.628 \text{ kg m}^2\text{s}^{-1}$$

Hence the angular momentum of the circular ring becomes $0.628 \text{ kg m}^2\text{s}^{-1}$.

Question 16

In an isothermal expansion of the gas the final volume becomes double of the initial volume, then which of the following statement is correct:

Options:

- A. The final pressure will be double of the initial pressure
- B. The final pressure will be half of the initial pressure
- C. The final pressure will be equal to the initial pressure
- D. None of these

Answer: B

Solution:**CONCEPT:****Isothermal process:**

- If the thermodynamic process is carried out at constant temperature then it is called an isothermal process.
- The change in internal energy in this process remains zero.
- The heat absorbed by the gas goes entirely to do work.

Boyle's law:

- According to this law if the temperature of a gas remains constant, then the pressure (P) exerted by a gas is inversely proportional to the volume (V) occupied by it.

$$\Rightarrow P \propto \frac{1}{V}$$

$$\Rightarrow PV = \text{constant}$$

CALCULATION:

Given Temperature = constant, $V_1 = V$ and $V_2 = 2V$

By Boyle's law,

$$\Rightarrow P_1 V_1 = P_2 V_2$$

$$\Rightarrow P_1 V = P_2 \times 2V$$

$$\Rightarrow P_1 = 2P_2$$

$$\Rightarrow P_2 = \frac{P_1}{2}$$

Hence, option 2 is correct.

Question 17

The magnetic flux through a 50-turn coil increases at the rate of 0.05 Wb/s. What is the induced emf between the ends of the coil?

Options:

A.

2.5 V

B.

5 V

C.

10 V

D.

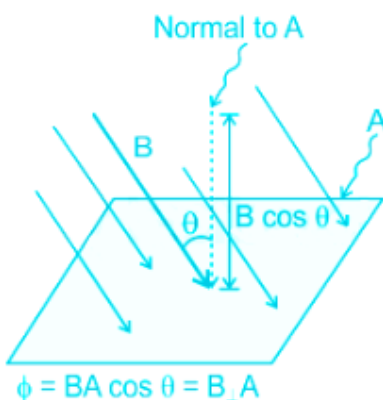
25 V

Answer: A

Solution:

CONCEPT:

- **Magnetic flux** is total magnetic field that passes through the given area.
- If we choose a simple flat surface with area **A** and there is an angle θ between the normal to the surface and a magnetic field vector (magnitude **B**) then the magnetic flux is $\phi = BA \cos \theta$



- **Faraday's Law:** Any change in the magnetic flux of wire will cause a voltage (**induced emf**) to be "induced" in the coil.
- This change can be produced by changing the magnetic field strength, moving the coil into or out of the magnetic field, moving a magnet toward or away from the coil, rotating the coil relative to the magnet, etc.
- **Induced emf** = $-N \frac{\Delta \phi}{\Delta t}$

CALCULATION:

Given that $N=50$ turns

$$\frac{\Delta\phi}{\Delta t} = 0.05 \text{ Wb/s}$$

$$\text{induced emf} = -N \frac{\Delta\phi}{\Delta t}$$

$$= 50 \times 0.05 = \mathbf{2.5 \text{ Volt}}$$

So, The correct answer is Option 1, i.e. **2.5 V**



Important Point

- **Lenz's law** says that the direction of the induced current in coil will be always in such a way as to oppose the change which produces the current.
- It is just a small addition to Faraday's law. See negative sign in the formula. Negative sign shows opposition.

Question 18

The condition for a thermodynamic process to be adiabatic:

Options:

A. $\Delta Q = 0$

B. $\Delta T = 0$

C. $\Delta V = 0$

D. $\Delta W = 0$

Answer: A

Solution:

CONCEPT:

Isobaric process	Isochoric process	Adiabatic process	Isothermal process
It allows to set up the relationship between the changes in volume and temperature under constant pressure .	The process in which the volume of the gas remains constant is called the isochoric process. For example: A gas is filled in a closed container then the volume of the gas will remain constant.	The thermodynamic process in a system, during which no heat transfer occurs between thermodynamic systems and surrounding is called an adiabatic process i.e., $\Delta Q = 0$	It allows to set up the relationship between the changes in pressure and volume under constant temperature :
$V_1/T_1 = V_2/T_2$ so $V \propto T$ Where V_1 and V_2 are volume and T_1 and T_2 are different	$\frac{P_1}{T_1} = \frac{P_2}{T_2} = \text{Constant}$	$PV^\gamma = \text{Constant}$ Where γ is ratio of specific heat	$P_1 V_1 = P_2 V_2$ so $P V = \text{Constant}$

temperatures]

Where $[P_1$ and P_2 are the pressure of gases and V_1 and V_2 are volume]

EXPLANATION:

- In the adiabatic process, the heat remains constant i.e., $\Delta Q = 0$. Therefore option 1 is correct.
-

Question 19

A 700 gm solid cube having an edge of length 10 cm floats in water. How much volume of the cube is outside the water? The density of the water 1000 kgm^{-3}

Options:

A. 300 cm^3

B. 100 cm^3

C. 400 cm^3

D. 200 cm^3

Answer: A

Solution:

Concept:

Archimedes Principle: When a body is fully or partially dipped in fluid, the fluid exerts a contact force on the body. The resultant of this contact force is known as buoyancy or buoyant force.

- This force acts vertically upwards (opposite to the weight of the body).
- Mathematically, $F = V\rho g$ [where, V = volume of liquid displaced, ρ = density of the liquid, g = acceleration due to gravity.]
- A body floats on a fluid if the average density of the body is less than that of fluid.
- The weight of the liquid displaced by the immersed part of the body must be equal to the weight of the body.

Calculation:

Given:

$m = 700 \text{ g} = 0.7 \text{ kg}$, $a = 10 \text{ cm}$

In floating bodies,

Weight of fluid = Buoyant force

Buoyant force = weight of water displaced. If V is the volume of cube inside the water, then the weight of water displaced = $V\rho g$.

$$\Rightarrow V\rho g = 0.7g$$

$$\Rightarrow V = \frac{0.7g}{\rho g} = \frac{0.7}{1000} = 7 \times 10^{-4} \text{ m}^3 = 700 \text{ cm}^3$$

$$\text{Total volume of the cube} = (10 \text{ cm})^3 = 1000 \text{ cm}^3$$

$$\therefore \text{Volume outside the water} = 1000 - 700 = 300 \text{ cm}^3$$

Question 20

An electric bulb rated 200 V and 100 W is connected to a 160 V power supply. What power will be consumed by the bulb?

Options:

A. 64 W

B. 80 W

C. 100 W

D. 160 W

Answer: A

Solution:

Concept:

Electric Power -

- Electric power is the rate at which electrical energy is transferred by an electric circuit.
- The SI unit of Power is the Watt or One Joule per second.
- Power in the resistor is given by
- $P = \frac{V^2}{R}$

Calculation:

Let the resistance of the bulb be R

Now since the bulb is rated at Power 100 W and Voltage 200 V

So, from equation 1

$$100 = \frac{200^2}{R}$$

$$\Rightarrow R = 400\Omega$$

When the bulb is connected to a 160 V power supply then power is

$$P = \frac{V^2}{R}$$

$$P = \frac{160^2}{400} = 64 \text{ W}$$

Question 21

110 joule of heat is added to a gaseous system, whose internal energy is 40 J. Then the amount of external work done is

Options:

A. 150 J

B. 70 J

C. 110 J

D. 40 J

Answer: B

Solution:

Concept:

- **The first law of thermodynamics is a restatement of the law of conservation of energy.** It states that energy cannot be created or destroyed in an isolated system; energy can only be transferred or changed from one form to another.
- When heat energy is supplied to a thermodynamic system or any machine.
- Two things may occur:
 - The **internal energy** of the system or machine may change.
 - The system may do some **external work**.

According to the **first law of Thermodynamics**:

$$\Delta Q = \Delta W + \Delta U$$

Where ΔQ = Heat supplied to the system or **heat exchange**, ΔW = work done by the system, ΔU = change in internal energy of the system

Explanation:

Given that, $\Delta Q = 110 \text{ J}$, $\Delta U = 40 \text{ J}$

According to the **first law of thermodynamics**:

$$\Delta Q = \Delta W + \Delta U$$

$$\Delta U = \Delta Q - \Delta W$$

$$40 \text{ J} = 110 \text{ J} - \Delta W$$

$$\Delta W = 110 - 40 = 70 \text{ J}$$

Then the amount of external work done is 70 J.

Question 22

The temperature of a gas is increased by 27°C . Then what is the equivalent increase on Kelvin scale is

Options:

A. 27 K

B. 273.9 K

C. 102 K

D. 30 K

Answer: A

Solution:

The correct answer is option 1) i.e. 27 K

CONCEPT:

- **Celsius scale** is based on the freezing point of water at 0°C and the boiling point of water at 100°C .
 - The region between these two temperatures is divided into **100 equal parts** i.e. each part equals 1°C .
 - The absolute zero value is at **-273.15°C** .
- **Kelvin scale** is a temperature scale that is based on the freezing point of water at **273 K** and the boiling point of water at **373 K**.
 - The region between these two temperatures is divided into **100 equal parts** i.e. each part equals **1 K**.
 - The absolute zero value is at **0 K**.

Temperature conversion formulas:

	to Celsius	to Kelvin
Celsius	$^\circ\text{C}$	$^\circ\text{C} + 273.15$
Kelvin	$\text{K} - 273.15$	K

EXPLANATION:

Given - Temperature change (ΔT) = 27°C

- Let T_1 and T_2 be the two temperatures in degrees Celsius. On Celsius scale:

$$\Rightarrow \Delta T = (T_2 - T_1)^\circ\text{C}$$

- On Kelvin scale:

$$\Rightarrow \Delta T = (T_2 + 273.15) - (T_1 + 273.15) = (T_2 - T_1) \text{ K}$$

- Therefore, the equivalent increase on Kelvin scale is 27 K.

Important Points

- The change in temperature ΔT will have the same numerical value on both the Celsius and Kelvin scale.
-

Question 23

If the refractive index of water is $\frac{4}{3}$ and that of glass is $\frac{5}{3}$, then the critical angle of incidence for light tending to go from glass to water is:

Options:

A.

$$\sin^{-1}\left(\frac{3}{4}\right)$$

B.

$$\sin^{-1}\left(\frac{3}{5}\right)$$

C.

$$\sin^{-1}\left(\frac{4}{5}\right)$$

D.

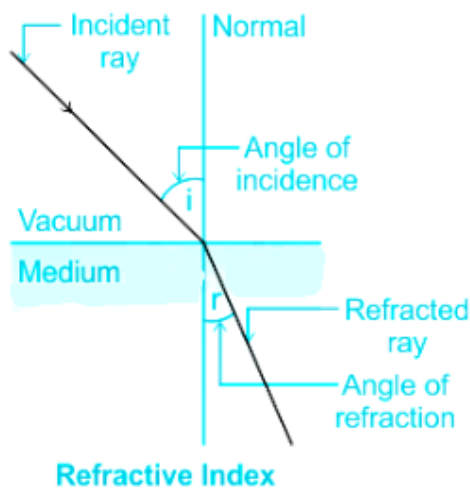
$$\sin^{-1}\left(\frac{2}{3}\right)$$

Answer: C

Solution:

Concept:

- The Refractive index also called the index of refraction measures the bending of a ray of light when passing from one medium into another.



- If i is the angle of incidence of a ray in vacuum (the angle between the incoming ray and the perpendicular to the surface of a medium, called the normal) and r is the angle of refraction (the angle between the ray in the medium and the normal),
- Then the refractive index n is defined as the ratio of the sine of the angle of incidence to the sine of the angle of refraction.
- $\mu = \frac{\sin i}{\sin r}$
- Refractive index of combined media = $\frac{\mu_1}{\mu_2}$ --- (1)
- Critical angle, $C = \sin^{-1}(\mu^{-1})$ --- (2)

Calculation:

Given, the refractive index of water, ${}_w\mu_a = \frac{4}{3}$

Refractive index of a glass slab, ${}_g\mu_a = \frac{5}{3}$

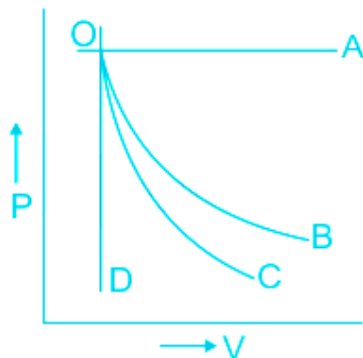
Refractive index of combined media = $\frac{{}_g\mu_a}{{}_w\mu_a} = \frac{5}{3} \times \frac{3}{4}$

From equation 1,

$$\mu = \frac{5}{4}$$

From equation 2 Critical angle, $C = \sin^{-1}\left(\frac{4}{5}\right)$

Question 24



A graph of pressure versus volume for an ideal gas for different processes is as shown. In the graph curve OC represents

Options:

- A. isothermal process
- B. isobaric process
- C. adiabatic process
- D. isochoric process

Answer: C

Solution:

CONCEPT:

- **Isothermal:** The thermodynamic process in a system, during which the **temperature remains constant** is called isothermal process.

$$\Delta T = 0$$

$$PV = nRT$$

$$PV = \text{const.}$$

- **Isochoric:** The thermodynamic process in a system, during which the **Volume remains constant** is called the isochoric process.

$$\Delta V = 0$$

- **Isobaric:** The thermodynamic process in a system, during which the pressure remains constant is called the Isobaric process.

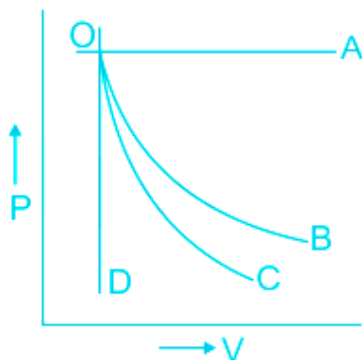
$$\Delta P = 0$$

- **Adiabatic:** The thermodynamic process in a system, during which **no heat or mass transfer occurs between thermodynamic systems** is called an adiabatic process.

$$\Delta Q = 0$$

$$PV^\gamma = \text{const.}$$

EXPLANATION:



Curve OA: $P = \text{const.}$ So the **Isobaric** process.

Curve OD: $V = \text{const.}$ So the **Isochoric** process.

Isothermal Process $PV = \text{const.}$ and adiabatic process $PV^\gamma = \text{const.}$, both represent **rectangular hyperbola**.

Curve OB and Curve OC:

In both the processes of expansion and compression, the **adiabatic curve is steeper than the isothermal curve**.

$$\left| \frac{\Delta P}{\Delta V} \right|_{adi} = -\gamma \left| \frac{\Delta P}{\Delta V} \right|_{iso}$$

- So the **slope of the adiabatic curve is more than the isothermal curve**.
- That's why the **Curve OB** is an **isothermal process**.
- **Curve OC is the adiabatic process**.
- Hence the correct answer is **option 3**.

Question 25

The magnitude of the magnetic field at a far distance along the axis of the solenoid is B . If the current in the solenoid is doubled then the magnetic field at that point will become:

Options:

- A. B
- B. $2B$
- C. $4B$
- D. None of these

Answer: B

Solution:

CONCEPT:

Solenoid:

- The solenoid consists of a long wire wound in the form of a helix where the neighbouring turns are closely spaced, so each turn can be regarded as a circular loop.
- The net magnetic field due to a solenoid is the vector sum of the fields due to all the turns.
- Enamelled wires are used for winding so that turns are insulated from each other.

The magnetic field at a distance on the axis of the solenoid:

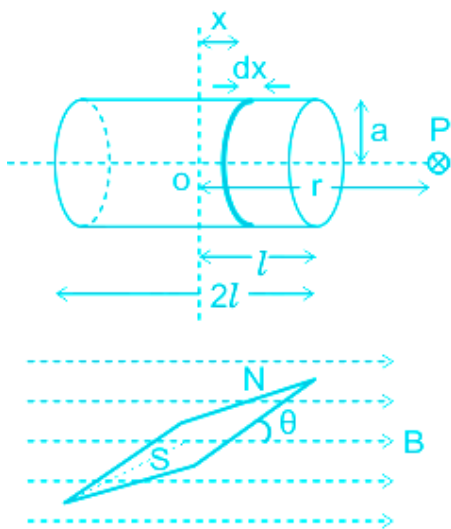
- The magnetic field on the axis at a far distance 'x' from the center of the solenoid is given as,

$$\Rightarrow B = \frac{\mu_0 n I}{2} \times \frac{2la^2}{r^3}$$

- The magnitude of the magnetic moment of the solenoid is given as,

$$\Rightarrow m = n \times (2l) \times I \times (\pi a^2) = (\text{total number of turns} \times \text{current} \times \text{cross-sectional area})$$

Where n = number of turns per unit length, I = current in the solenoid, $2l$ = length of the solenoid, and a = radius of the solenoid



EXPLANATION:

Given B = magnitude of the magnetic field at a far distance along the axis of the solenoid

We know that the magnetic field on the axis at a far distance 'r' from the center of the solenoid is given as,

$$\Rightarrow B = \frac{\mu_0 n I}{2} \times \frac{2la^2}{r^3} \text{ ----(1)}$$

Where n = number of turns per unit length, I = current in the solenoid, $2l$ = length of the solenoid, and a = radius of the solenoid

If the current in the solenoid is doubled then the magnetic field at that point is given as ($I' = 2I$),

$$\Rightarrow B' = \frac{\mu_0 n I'}{2} \times \frac{2la^2}{r^3}$$

$$\Rightarrow B' = \frac{\mu_0 n (2I)}{2} \times \frac{2la^2}{r^3}$$

$$\Rightarrow B' = 2 \times \frac{\mu_0 n I}{2} \times \frac{2la^2}{r^3} \text{ ----(2)}$$

By equation 1 and equation 2,

$$\Rightarrow B' = 2B$$

- Hence, option 2 is correct.

Maths

Question 26

What is $\frac{2 \tan \theta}{1 + \tan^2 \theta}$ equal to?

Options:

A. $\cos 2\theta$

B. $\tan 2\theta$

C. $\sin 2\theta$

D. $\operatorname{cosec} 2\theta$

Answer: C

Solution:

Concept:

- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \tan^2 \theta = \sec^2 \theta$
- $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
- $2 \sin \theta \cos \theta = \sin 2\theta$

Calculation:

We have to find the value of $\frac{2 \tan \theta}{1 + \tan^2 \theta}$

$$\frac{2 \tan \theta}{1 + \tan^2 \theta} = \frac{2 \tan \theta}{\sec^2 \theta} = 2 \times \frac{\sin \theta}{\cos \theta} \times \cos^2 \theta = 2 \times \sin \theta \times \cos \theta = \sin 2\theta$$

$\therefore$ Option 3 is correct.

Question 27

The roots of the equation $3x^2 + 8x + 9 = 0$ are α, β , then equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$ is

Options:

A. $3x^2 - 8x + 9 = 0$

B. $9x^2 - 8x + 3 = 0$

C. $9x^2 + 8x + 3 = 0$

D. $3y^2 + 9y + 8 = 0$

Answer: C

Solution:

Concept:

General Form of Quadratic Equation, $ax^2 + bx + c = 0$

1. Sum of roots, $\alpha + \beta = \frac{-b}{a}$

2. Products of roots, $\alpha\beta = \frac{c}{a}$

3. The quadratic equation whose roots are α and β is given by $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

Calculation:

Given

$$\Rightarrow 3x^2 + 8x + 9 = 0$$

$$\Rightarrow \text{sum of roots } \alpha + \beta = -\frac{8}{3}$$

$$\Rightarrow \text{Products of roots } \alpha\beta = \frac{9}{3} = 3$$

Given two roots are $\frac{1}{\alpha}, \frac{1}{\beta}$

$$\text{Sum of roots} = \frac{1}{\alpha} + \frac{1}{\beta}$$

$$\text{Products of roots} = \frac{1}{\alpha} \cdot \frac{1}{\beta}$$

The quadratic equation is

$$\Rightarrow x^2 - x(\text{Sum of roots}) + \text{Products of roots} = 0$$

$$\Rightarrow x^2 - x\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + \frac{1}{\alpha} \cdot \frac{1}{\beta} = 0$$

$$\Rightarrow x^2 - x\left(\frac{\alpha + \beta}{\alpha\beta}\right) + \frac{1}{\alpha\beta} = 0$$

$$\Rightarrow x^2 - x\left(-\frac{8}{9}\right) + \frac{1}{3} = 0$$

$$\Rightarrow 9x^2 + 8x + 3 = 0$$

★ Important Points

Relation between Roots and Coefficients

- If the roots of quadratic equation $ax^2 + bx + c$, $a \neq 0$, are α and β , then $\alpha + \beta = \frac{-b}{a} = -\frac{\text{Coefficient of } x}{\text{Coefficient of } x^2}$ and $\alpha\beta = \frac{c}{a} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$
 - The quadratic equation whose roots are α and β is given by $x^2 - (\alpha + \beta)x + \alpha\beta = 0$
-

Question 28

One card is drawn from a well shuffled pack of 52 cards. Find the probability that it is a king.

Options:

A. $\frac{1}{4}$

B. $\frac{1}{52}$

C. $\frac{1}{26}$

D. None of these.

Answer: D

Solution:

Concept:

The probability of the occurrence of an event A out of a total possible outcomes N, is given by: $P(A) = \frac{n(A)}{N}$, where $n(A)$ is the number of ways in which the event A can occur.

Calculation:

In a pack of 52 cards, there are 4 kings.

The number of ways of getting one king, if one card is drawn, will be $n(A) = {}^4C_1 = 4$.

Total number of ways of drawing one cards is $N = {}^{52}C_1 = 52$.

The required probability $= \frac{n(A)}{N} = \frac{4}{52} = \frac{1}{13}$.

Question 29

What is the equation of a straight line which passes through (3, 4) and sum of the whose x and y intercepts is 14?

Options:

A. $4x + 3y = 24$

B. $x + y = 14$

C. $4x - 3y = 0$

D. $3x + 4y = 25$

Answer: A

Solution:

Concept:

The equation of line in intercept form is $\frac{x}{a} + \frac{y}{b} = 1$

Calculations:

Let x - intercept = a and y - intercept = b

Given, sum of x and y - intercept is 14.

$$\Rightarrow a + b = 14.$$

We know that equation of line in intercept form is $\frac{x}{a} + \frac{y}{b} = 1$

Let the equation of a straight line which passes through (3, 4) and sum of the whose x and y intercepts is 14 is

$$\frac{3}{a} + \frac{4}{14-a} = 1$$

$$\Rightarrow \frac{3(14-a)+4a}{a \times (14-a)} = 1$$

$$\Rightarrow 3(14 - a) + 4a = a(14 - a)$$

$$\Rightarrow a^2 - 13a + 42 = 0$$

$$\Rightarrow (a - 7)(a - 6) = 0$$

$$\Rightarrow a = 7 \text{ or } a = 6$$

If $a = 7$ then $b = 7$ and if $a = 6$ then $b = 8$

So, equation of line is $\frac{x}{6} + \frac{y}{8} = 1$ or $\frac{x}{7} + \frac{y}{7} = 1$

So equation of line $4x + 3y = 24$ or $x + y = 7$.

Question 30

If $\vec{a} = (\vec{i} + 2\vec{j} - 3\vec{k})$ and $\vec{b} = (3\vec{i} - \vec{j} + 2\vec{k})$ then the angle between $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ is?

Options:

A. $\pi / 3$

B. $\pi / 4$

C. $\pi / 2$

D. $2\pi / 3$

Answer: C

Solution:

Concept:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

Calculation:

$$\text{Here, } \vec{a} = (\vec{i} + 2\vec{j} - 3\vec{k}) \text{ and } \vec{b} = (3\vec{i} - \vec{j} + 2\vec{k})$$

$$(\vec{a} + \vec{b}) = (\vec{i} + 2\vec{j} - 3\vec{k}) + (3\vec{i} - \vec{j} + 2\vec{k}) = 4\vec{i} + \vec{j} - \vec{k} \quad |\vec{a} + \vec{b}| = \sqrt{4^2 + 1^2 + (-1)^2} = \sqrt{18} = 3\sqrt{2}$$

$$(\vec{a} - \vec{b}) = (\vec{i} + 2\vec{j} - 3\vec{k}) - (3\vec{i} - \vec{j} + 2\vec{k}) = -2\vec{i} + 3\vec{j} - 5\vec{k} \quad |\vec{a} - \vec{b}| = \sqrt{(-2)^2 + 3^2 + (-5)^2} = \sqrt{38}$$

Now,

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = (4\vec{i} + \vec{j} - \vec{k}) \cdot (-2\vec{i} + 3\vec{j} - 5\vec{k}) = -8 + 3 + 5 = 0 \quad |(\vec{a} + \vec{b})| |(\vec{a} - \vec{b})| \cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

Hence, option (3) is correct.

Question 31

The value of $\tan \left(\cos^{-1} \frac{3}{5} + \tan^{-1} \frac{1}{4} \right)$ is

Options:

A. $\frac{19}{8}$

B. $\frac{8}{19}$

C. $\frac{19}{12}$

D. $\frac{3}{4}$

Answer: A**Solution:****Explanation:**

$$\Rightarrow \text{We know, } \tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right) \forall xy < 1$$

$$\text{We have, } \tan\left(\cos^{-1}\frac{3}{5} + \tan^{-1}\frac{1}{4}\right)$$

$$\text{Let } \cos^{-1}\left(\frac{3}{5}\right) = \theta$$

$$\Rightarrow \cos\theta = 3/5$$

$$\Rightarrow \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\sqrt{1-(\frac{3}{5})^2}}{\frac{3}{5}} = 4/3$$

$$\Rightarrow \theta = \tan^{-1}(4/3)$$

$$\therefore \tan\left(\cos^{-1}\frac{3}{5} + \tan^{-1}\frac{1}{4}\right)$$

$$= \tan\left(\theta + \tan^{-1}\frac{1}{4}\right)$$

$$= \tan\left(\tan^{-1}\frac{4}{3} + \tan^{-1}\frac{1}{4}\right)$$

$$= \tan\left(\tan^{-1}\left(\frac{\frac{4}{3} + \frac{1}{4}}{1 - \frac{4}{3} \times \frac{1}{4}}\right)\right)$$

$$= \tan\left(\tan^{-1}\left(\frac{19}{8}\right)\right)$$

$$= 19/8$$

$$\therefore \tan\left(\cos^{-1}\frac{3}{5} + \tan^{-1}\frac{1}{4}\right) = 19/8$$

Question 32

Find the length of the latus rectum of the parabola $y^2 = -12x$?

Options:

- A. 8
- B. 12
- C. 10
- D. None of these

Answer: B

Solution:

CONCEPT:

The following are the properties of a parabola of the form: $y^2 = -4ax$ where $a > 0$

- Focus is given by $(-a, 0)$
- Vertex is given by $(0, 0)$
- Equation of directrix is given by: $x = a$
- Equation of axis is given by: $y = 0$
- Length of latus rectum is given by: $4a$
- Equation of latus rectum is given by: $x = -a$

CALCULATION:

Given: Equation of parabola is $y^2 = -12x$

The given equation can be re-written as: $y^2 = -4 \cdot 3 \cdot x$ -----(1)

Now by comparing the equation (1), with $y^2 = -4ax$ we get

$$\Rightarrow a = 3$$

As we know that, the length of latus rectum of a parabola is given by: $4a$

So, length of latus rectum of the given parabola is: $4 \cdot 3 = 12$ units

Hence, **option B** is the correct answer.

Question 33

The value of $\int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan^3 x}$ is

Options:

- A. 0

B. 1

C. $\pi/2$

D. $\pi/4$

Answer: D

Solution:

Concept:

- $\tan x = \frac{\sin x}{\cos x}$
- $\sin\left(\frac{\pi}{2} - x\right) = \cos x$
- $\cos\left(\frac{\pi}{2} - x\right) = \sin x$
- $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

Calculation:

Given,

$$I = \int_0^{\frac{\pi}{2}} \frac{dx}{1 + \tan^3 x}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{dx}{1 + \frac{\sin^3 x}{\cos^3 x}}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\cos^3 x dx}{\cos^3 x + \sin^3 x} \text{ ---- (i)}$$

By using property,

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\cos^3\left(\frac{\pi}{2} - x\right) dx}{\cos^3\left(\frac{\pi}{2} - x\right) + \sin^3\left(\frac{\pi}{2} - x\right)}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\sin^3 x dx}{\sin^3 x + \cos^3 x} \text{ ---- (ii)}$$

Adding (i) and (ii),

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \frac{\sin^3 x + \cos^3 x}{\sin^3 x + \cos^3 x} dx$$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} dx$$

$$\Rightarrow I = \frac{\pi}{4}$$

$\therefore$ The correct answer is option (4).

Question 34

$$\lim_{x \rightarrow 0} \frac{\sin(2+x) - \sin(2-x)}{x} = ?$$

Options:

A. $\frac{1}{2} \cos 2$

B. 1

C. $2 \cos 2$

D. 9

Answer: C

Solution:

Concept:

$$\sin(a) - \sin(b) = 2 \cos\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right) \dots (1)$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \dots (2)$$

Explanation:

We are given $\lim_{x \rightarrow 0} \frac{\sin(2+x) - \sin(2-x)}{x}$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2 \cos\left(\frac{(2+x)+(2-x)}{2}\right) \sin\left(\frac{(2+x)-(2-x)}{2}\right)}{x} \text{ (Using equation 1)}$$

$$\Rightarrow \lim_{x \rightarrow 0} 2 \cos(2) \frac{\sin(x)}{x}$$

$$\Rightarrow 2 \cos(2) \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$\Rightarrow 2 \cos(2) \times 1 \text{ (Using equation 2)}$$

$$\Rightarrow 2 \cos 2$$

Question 35

In the expansion of $(1 + x)^{50}$ the sum of the coefficients of odd powers of x is:

Options:

- A. 0
- B. 2^{49}
- C. 2^{50}
- D. 2^{51}

Answer: B

Solution:

Concept:

We can expand a polynomial using the binomial expansion formula as follows:

$$(1 + x)^n = {}^nC_0 x^0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

$$\text{Where, } {}^nC_r = \frac{n!}{r!(n-r)!}$$

Calculation:

We know that,

$$(1 + x)^n = {}^nC_0 x^0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

$$\Rightarrow (1 + x)^{50} = C_0 + C_1 x + C_2 x^2 + \dots + C_{50} x^{50}$$

Putting $x = 1$ & $x = -1$ we will get

$$2^{50} = C_0 + C_1 + C_2 + C_3 + \dots + C_{50} \text{ ----(1)}$$

$$0 = C_0 - C_1 + C_2 - C_3 + \dots + C_{50} \text{ ----(2)}$$

Subtracting equation (2) from equation(1), we get

$$2^{50} - 0 = 2[C_1 + C_3 + C_5 + \dots]$$

$$\Rightarrow C_1 + C_3 + C_5 + \dots = 2^{49}$$

Hence, sum of coefficients of odd powers of $x = 2^{49}$.

Question 36

$\tan \theta = \frac{-3}{4}$ and θ lies in the second quadrant, then $\sin \theta$ is equal to

Options:

A. $\frac{2}{5}$

B. $\frac{2}{3}$

C. $\frac{3}{5}$

D. $\frac{4}{5}$

Answer: C

Solution:



Hint

- $\sin \theta = \frac{\text{Opposite Side}}{\text{Hypotenuse}}$
- $\tan \theta = \frac{\text{Opposite Side}}{\text{Adjacent Side}}$

Explanation:

$\tan \theta = \frac{-3}{4}$ and θ lies in the second quadrant.

But we know, $\sin \theta$ is positive in the second quadrant.

Also, by Pythagorean triplet, hypotenuse = 5

$\Rightarrow \sin \theta = \frac{3}{5}$

Question 37

The radius of the circle $x^2 + y^2 + x + c = 0$ passing through the origin is

Options:

A. $1/4$

B. $1/2$

C. 1

D. 2

Answer: B

Solution:

Concept:

Let $x^2 + y^2 = r^2$ is the equation of circle. then (0, 0) is the origin and r is the radius of the circle.

Calculations:

We know that, $x^2 + y^2 = r^2$ is the equation of circle. then (0, 0) is the origin and r is the radius of the circle.

Given equation of circle is $x^2 + y^2 + x + c = 0$, which is passing through the origin.

i.e. $c = 0$.

$$\Rightarrow x^2 + y^2 + x = 0.$$

$$\Rightarrow x^2 + x + \frac{1}{4} - \frac{1}{4} + y^2 = 0$$

$$\Rightarrow x^2 + x + \frac{1}{4} + y^2 = \frac{1}{4}$$

$$\Rightarrow \left(x + \frac{1}{2}\right)^2 + y^2 = \left(\frac{1}{2}\right)^2$$

which is equation of circle with radius is $\frac{1}{2}$.

Hence, the radius of the circle $x^2 + y^2 + x + c = 0$ passing through the origin is $\frac{1}{2}$.

Question 38

$$\int_{-1}^1 \frac{2x+1}{(x^2+x+1)^2} dx =$$

Options:

A. $-\frac{1}{3}$

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. $-\frac{2}{3}$

Answer: C

Solution:

Concept:

Integral property:

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C; n \neq -1$
- $\int \frac{1}{x} dx = \ln x + C$
- $\int e^x dx = e^x + C$

Calculation:

$$I = \int \frac{2x+1}{(x^2+x+1)^2} dx$$

$$\text{Let } x^2 + x + 1 = t$$

$$\Rightarrow (2x + 1) dx = dt$$

$$I = \int \frac{dt}{t^2}$$

$$I = \frac{t^{-1}}{-1}$$

$$I = \frac{-1}{t}$$

$$I = \frac{-1}{x^2+x+1}$$

Putting limits

$$I = \left[\frac{-1}{x^2+x+1} \right]_{-1}^1$$

$$I = \frac{-1}{1^2+1+1} - \left(\frac{-1}{(-1)^2+(-1)+1} \right)$$

$$I = 1 - \frac{1}{3} = \frac{2}{3}$$

Question 39

The value of $i^3 + i^6 + i^9$, where $i = \sqrt{-1}$, is

Options:

A. 1

B. -1

C. i

D. -i

Answer: B

Solution:

Concept:

Power of i:

- $i = \sqrt{-1}$
- $i^2 = -1$
- $i^3 = i \times i^2 = -i$
- $i^4 = (i^2)^2 = (-1)^2 = 1$
- $i^{4n} = 1$

Calculation:

To Find: Value of $i^3 + i^6 + i^9$

As we know, $i^2 = -1$

So, $i^3 = i^2 \times i = -1 \times i = -i$

$i^6 = i^3 \times i^3 = -i \times -i = i^2 = -1$

$i^9 = i^3 \times i^3 \times i^3 = -i \times -i \times -i = -i^3 = -(-i) = i$

Now,

$i^3 + i^6 + i^9 = (-i) + (-1) + i = -1$

Question 40

$$\int \frac{1}{e^x + e^{-x}} dx =$$

Options:

A. $\log |\cot(e^x) + \tan(e^x)|$

B. $\sin^{-1}(e^x) + c$

C. $\log |1 + e^x|$

D. $\tan^{-1}(e^x) + c$

Answer: D

Solution:

Concept:

$$\int \frac{1}{1+x^2} dx = \tan^{-1}x + c$$

$$x^{-1} = \frac{1}{x}$$

Calculation:

$$\text{Let, } I = \int \frac{1}{e^x + e^{-x}} dx$$

$$= \int \frac{1}{e^x + \frac{1}{e^x}} dx \quad (\because x^{-1} = \frac{1}{x})$$

$$= \int \frac{e^x}{e^{2x} + 1} dx$$

Now, let $e^x = t$

$$\Rightarrow e^x dx = dt$$

$$\therefore I = \int \frac{dt}{t^2 + 1}$$

$$= \tan^{-1}t + c \quad (\because \int \frac{1}{1+x^2} dx = \tan^{-1}x + c)$$

$$= \tan^{-1}(e^x) + c \quad (\because e^x = t)$$

Hence, option (4) is correct.

Question 41

Average of all the natural numbers less than 150 is:

Options:

A. 70

B. 72

C. 75

D. 80

Answer: C

Solution:

Concept used:

Sum of first n natural numbers = $n(n+1)/2$

Calculation:

Sum of n natural numbers = $149 \times (150)/2$

$$\Rightarrow 149 \times 75$$

$$\Rightarrow 11175$$

$$\text{Average} = 11175/149 = 75$$

$\therefore$ The correct answer is 75.

Question 42

Determine k so that the line $y = 2x + k$ may touch the ellipse $3x^2 + 5y^2 = 15$

Options:

A. $\pm\sqrt{23}$

B. $\pm\sqrt{32}$

C. $\pm\sqrt{18}$

D. None of these

Answer: A

Solution:

Given:

$$y = 2x + k \text{ is line}$$

$$\text{Equation of ellipse } 3x^2 + 5y^2 = 15$$

Calculation:

$$\text{We have } y = 2x + k$$

Put the value of y in the equation of an ellipse, we get

$$3x^2 + 5(2x + k)^2 = 15$$

$$\Rightarrow 3x^2 + 20x^2 + 5k^2 + 20xk = 15$$

$$\Rightarrow 23x^2 + 20kx + 5k^2 - 15 = 0$$

For getting real value, $D > 0$

$$D = b^2 - 4ac$$

$$= (20k)^2 - 4 \times 23 \times (5k^2 - 15)$$

$$= -60k^2 + 1380$$

Now, $D > 0$

$$\Rightarrow -60k^2 + 1380 > 0$$

$$\Rightarrow 1380 > 60k^2$$

$$\Rightarrow k < \pm\sqrt{23}$$

Question 43

Find the 4th term from the end in the expansion $\left(\frac{x}{2} - x\right)^{15}$

Options:

A. $\frac{455x^{15}}{8}$

B. $\frac{45x^5}{8}$

C. $-\frac{425x^{15}}{8}$

D. More than one of the above

Answer: A

Solution:

Concept:

In the expansion $(x + y)^n$, pth term from the end is equal to $(n - p + 2)$ th term from the beginning.

$$\text{General Term} = T_{r+1} = {}^nC_r x^{n-r} \cdot y^r$$

Calculation:

$$\text{Given: } \left(\frac{x}{2} - x\right)^{15}$$

In the given expansion 4th term from the end is equal to $(15 - 4 + 2)$ th term from the beginning (i.e., 13th term)

$$\therefore 4\text{th term from the end} = T_{13} = T_{12+1}$$

$$T_{12+1} = (-1)^{12} \times {}^{15}C_{12} \times \left(\frac{x}{2}\right)^3 \times x^{12}$$

$$= {}^{15}C_3 \times \left(\frac{x^3}{8}\right) \times x^{12}$$

$$= \frac{15 \times 14 \times 13}{3 \times 2 \times 1} \times \left(\frac{x^{15}}{8}\right)$$

$$= \frac{455x^{15}}{8}$$

Hence, option (1) is correct.

Question 44

The locus of poles of normal chord to the parabola $y^2 = 4ax$ is

Options:

A. $(x + 2a)y^2 - 4a^3 = 0$

B. $(x + 2a)y^2 + 4a^3 = 0$

C. $(x - 2a)y^2 + 4a^3 = 0$

D. $(x - 2a)y^2 - 4a^3 = 0$

Answer: B

Solution:

Let the pole of the parabola $y^2 = 4ax$ be $P(n, k)$

Equation of normal chord is;

$$y = -tx + 2at + at^3$$

$$y + tx - 2at - at^3 = 0 \dots 1)$$

equation of the polar with respect to P is $T = 0$

$$ky = 2a(x + n)$$

$$2ax - ky + 2an = 0 \dots 2)$$

Now equation 1) & equation 2) represents the equation of same line

So, comparing both equations;

$$\frac{t}{2a} = \frac{1}{-k} = \frac{-2at - at^3}{2an}$$

$$\frac{t}{2a} = \frac{1}{-k} = \frac{-2t - t^3}{2n}$$

$$\frac{t}{2a} = \frac{1}{-k}$$

$$t = \frac{-2a}{k} \dots 3)$$

Also,

$$\frac{t}{2a} = \frac{-2t-t^3}{2n}$$

$$\frac{1}{a} = \frac{-2-t^2}{n}$$

$$\frac{n}{a} + 2 = -t^2$$

Substituting t from 3)

$$\frac{n+2a}{a} = -\left(-\frac{2a}{k}\right)^2$$

$$k^2 (n + 2a) = -4a^2$$

$$k^2 (n + 2a) + 4a^2 = 0$$

Generalizing the equation, we get;

$$y^2 (x + 2a) + 4a^2 = 0$$

Question 45

The inverse of a matrix A is given by $\begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$ What is A equal to?

Options:

A. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

B. $\begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$

D. $\begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$

Answer: A

Solution:

Concept:

$$AA^{-1} = I$$

Calculation:

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

We know that $AA^{-1} = I$

$$\therefore \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2a + \frac{3}{2}b & a - \frac{b}{2} \\ -2c + \frac{3d}{2} & c - \frac{d}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Two matrices are equal if their order and corresponding elements are equal.

$$\text{Therefore, } -2a + \frac{3}{2}b = 1 \text{ --- (1)}$$

$$\text{and } a - \frac{b}{2} = 0 \Rightarrow a = \frac{b}{2} \text{ --- (2)}$$

From equations (1) and (2),

$$\Rightarrow -2\left(\frac{b}{2}\right) + \frac{3}{2}b = 1$$

$$\Rightarrow -b + \frac{3b}{2} = 1$$

$$\Rightarrow \frac{b}{2} = 1$$

$$\therefore \mathbf{b = 2}$$

From equation (2), $\mathbf{a = 1}$

$$\text{Similarly, } -2c + \frac{3d}{2} = 0 \text{ --- (3)}$$

$$\text{and } c - \frac{d}{2} = 1 \text{ --- (4)}$$

Solving equation (3) and (4), we get $\mathbf{c = 3}$ and $\mathbf{d = 4}$

$$\therefore \text{Matrix } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Question 46

If -2 is the one of the roots of a quadratic equation $x^2 + 3x + 2 = 0$, then the other root is.

Options:

A. 2

B. -1

C. 1

D. 3

Answer: B

Solution:

Concept:

Sum and Product of Roots of a Quadratic Equation:

Let $ax^2 + bx + c = 0$ where $a \neq 0$

We can find the sum of the roots of this equation without determining the roots. The formula is given below;

$$\text{Sum of roots} = \frac{-b}{a}$$

Similarly we can find the product of the roots by using the following formula;

$$\text{Product of roots} = \frac{c}{a}$$

Solution:

Given: The equation is $x^2 + 3x + 2 = 0$

and -2 is the one of the roots of this equation.

Let the other root of this equation is k.

$$\text{Then, Sum of the roots} = \frac{-b}{a}$$

$$\Rightarrow -2 + k = \frac{-3}{1}$$

$$\Rightarrow k = -3 + 2$$

$$\Rightarrow k = -1$$

Hence, The other root of the equation is -1.

$\therefore$ The correct option is (2)

Question 47

In a G.P. tenth term is 9 and fourth term is 4, then its seventh term will be-

Options:

A. 8

B. 6

C. 9

D. 7

Answer: B

Solution:

Given:

In a G.P. tenth term is 9 and fourth term is 4,

Concept used:

$$n^{\text{th}} \text{ term of GP} = ar^{n-1}$$

Calculation:

$$a_{10} = 9, a_4 = 4$$

$$\therefore 9 = ar^9 \text{ \& } 4 = ar^3$$

$$\Rightarrow (ar^9)(ar^3) = 36$$

$$\Rightarrow a^2 r^{12} = 36$$

$$\Rightarrow (ar^6)^2 = (6)^2$$

Taking square root,

$$ar^6 = 6$$

$$\therefore 7^{\text{th}} \text{ term is } 6.$$

Question 48

Find the order and degree of the differential equation $(\frac{d^3 y}{dx^3})^2 = \sqrt{\frac{dy}{dx} + 1}$?

Options:

A. 3, 4

B. 4, 3

C. 2, 1

D. 1, 2

Answer: A

Solution:

Concept:

Order of differential equation: The order of highest order derivative involved in a differential equation is called the order of the differential equation.

Degree of differential equation: The highest degree of the highest order derivative involved in a differential equation is said to be the degree of the differential equation. Before finding a degree in terms of differential coefficient it must be free from radicals, fractions and functions.

Calculation:

$$\left(\frac{d^3y}{dx^3}\right)^2 = \sqrt{\frac{dy}{dx} + 1}$$

$$\Rightarrow \left(\frac{d^3y}{dx^3}\right)^4 = \frac{dy}{dx} + 1$$

$$\Rightarrow \left(\frac{d^3y}{dx^3}\right)^4 - \frac{dy}{dx} - 1 = 0$$

Order = 3 and Degree = 4

Question 49

What is $\lim_{x \rightarrow 4} \frac{3 - \sqrt{5+x}}{x-4}$ equal to?

Options:

A. $\frac{-1}{2}$

B. $\frac{-1}{6}$

C. $\frac{1}{6}$

D. $\frac{-1}{5}$

Answer: B

Solution:

Concept:

L-Hospital Rule: Let $f(x)$ and $g(x)$ be two functions

Suppose that we have one of the following cases,

$$\text{I. } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$$

$$\text{II. } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$$

Then we can apply L-Hospital Rule $\Leftrightarrow \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

Note: We have to differentiate both the numerator and denominator with respect to x unless and until

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = l \neq \frac{0}{0}$ where l is a finite value.

Solution:

We have to find the value of $\lim_{x \rightarrow 4} \frac{3 - \sqrt{5+x}}{x-4}$

$\lim_{x \rightarrow 4} \frac{3 - \sqrt{5+x}}{x-4}$ Form of limit is $(0/0)$

Applying L-Hospital rule,

$$= \lim_{x \rightarrow 4} \frac{0 - \frac{1}{2\sqrt{5+x}}}{1-0}$$

$$= \frac{-1}{2\sqrt{9}}$$

$$= \frac{-1}{6}$$

$\therefore$ The correct option is (2)

Question 50

Argument and modulus of $\frac{1+i}{1-i}$ are respectively

Options:

A. $\frac{-\pi}{2}$ and 1

B. $\frac{\pi}{2}$ and $\sqrt{2}$

C. 0 and $\sqrt{2}$

D. $\frac{\pi}{2}$ and 1

Answer: D

Solution:

Concept

The general form of a complex number is $z = x + iy$. The polar representation of z

is $z = r(\cos \theta + i \sin \theta)$. Here, r is the modulus of z and θ is called the amplitude or argument of the complex number. The formula to find the amplitude of a complex

number is: $\theta = \tan^{-1} \frac{y}{x}$ and $|z| = \sqrt{x^2 + y^2}$

Calculation

$$\frac{1+i}{1-i}$$

$$\Rightarrow \frac{1+i}{1-i} = \frac{1+i}{1-i} \times \frac{1+i}{1+i}$$

$$\Rightarrow \frac{1+i}{1-i} = \frac{(1+i)^2}{1-i^2} = \frac{(1+i)^2}{2} = \frac{1+i^2+2i}{2} = i$$

$$\Rightarrow \frac{1+i}{1-i} = 0 + i$$

$$\text{Now, Argument } = \theta = \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{i}{0} = \frac{\pi}{2}$$

$$\text{Modulus is } |z| = \sqrt{x^2 + y^2} = \sqrt{0^2 + 1^2} = 1$$

$\therefore$ Argument and modulus of $\frac{1+i}{1-i}$ are respectively $\frac{\pi}{2}$ and 1.

English

Question 51

Some parts of the sentence have errors and some are correct. Find out which part has an error and mark that part as your answer. If there are no errors, mark 'No error' as your answer.

Hardly had he departed that the phone rang.

Options:

- A. Hardly had
- B. he departed that
- C. the phone rang
- D. No error

Answer: B

Solution:

The erroneous part is "he departed that"



Key Points

- In Option 2, the usage of "that" is incorrect instead of this we need to use "when".
 - With "**hardly/scarcely**" we always use a conjunction "**when**".
 - Let's see an example-
 - **Hardly** had I reached the station **when** the train came.
 - Hence, the correct sentence is- *Hardly had he departed **when** the phone rang.*
-

Question 52

Identify the part of the sentence which has an error in it. In case there is no error the answer will be (D).

Ram has an (A)/ large amount(B)/ of money. (C)/ No error (D)

Options:

- A. A
- B. B
- C. C
- D. D

Answer: A

Solution:

The correct answer is "A".

Key Points

- In the given sentence, the usage of "an" is incorrect. Instead of "an", we need to use "a".
- We use "an" with nouns which have vowel sound and we use "a" with nouns which have consonant sound. "large" has consonant sound. Hence, we need to use "a".
- The correct sentence is: "Ram has *a large amount of money*".

Additional Information

- *Although 'Number and amount' have similar meanings, there is a difference between them:*
 - 'Number' is used for things that can be counted, while 'Amount' is used for things that cannot be counted.
- *For example:*
 - The **number** of butter was not enough for him. 
 - The **amount** of butter was not enough for him. 

Question 53

Directions: Each item in this section has a sentence with three underlined parts labelled as (a), (b) and (c). Read each sentence to find out whether there is any error in any underlined part and indicate your response on the answer sheet against the corresponding letter, i.e., (a) or (b) or (c). If you find no error, your response should be indicated as (d).

They had to wait for the luggages (a)/ to be put into the cab (b)/ before they themselves got into it(c)/ No error(d).

Options:

- A. (a)
- B. (b)
- C. (c)
- D. (d)

Answer: A

Solution:

The correct answer is "They had to wait for the luggage".

Key Points

- There are some **uncountable nouns** (followed by a singular verb) which don't take the articles 'a/an', many, few, and plural forms:
 - Furniture, scenery, luggage, temper, hair, knowledge, equipment, pottery, music, jewellery, etc.
 - Example:
 - *The **sceneries** at my friend's house enchanted me.* ❌
 - *The **scenery** at my friend's house enchanted me.* ✅
- As per the rule given above, '**luggage**' will be used instead of '**luggages**' in the 4th part of the given question.

Correct sentence: *They had to wait for the **luggage** to be put into the cab before they themselves got into it.*

Additional Information

- The nouns given below seem plural in form but in actuality are singular:
 - news, billiards, summons, physics, measles, innings, etc.
 - Example:
 - ***Physics** is an interesting subject.*
- हॉ कुछ **uncountable nouns** (followed by a singular verb) जो article 'a/an', many, few, and plural form: नहीं लेते हैं:
 - Furniture, scenery, luggage, temper, hair, knowledge, equipment, pottery, music, jewellery, etc.
 - उदाहरण:
 - *The **sceneries** at my friend's house enchanted me.* ❌
 - *The **scenery** at my friend's house enchanted me.* ✅
- ऊपर दिए गए नियम के अनुसार, '**luggage**' को देखते हुए सवाल के 4th part में '**luggages**' के स्थान पर इस्तेमाल किया जाएगा।

सही वाक्य: *They had to wait for the **luggage** to be put into the cab before they themselves got into it.*

Question 54

Select the option that can be used as a one-word substitute for the given group of words.

A decision on which one cannot go back

Options:

- A. Irrevocable
- B. Improbable
- C. Incredible
- D. Incurable

Answer: A

Solution:

The correct answer is '**Irrevocable**'.

Key Points

- Let us look at the meaning of the given options -
 - **Irrevocable** -not able to be changed, reversed, or recovered; final.
 - **Improbable** -not likely to be true or to happen.
 - **Incredible** -impossible to believe.
 - **Incorrigible** -(of a person or their behaviour) not able to be changed or reformed.
 - From the above meanings it is clear that **irrevocable** is the best word to substitute the given group of words.
 - Hence, **option 1** is the correct answer.
-

Question 55

Choose the correctly spelt word.

Options:

- A. Peirce
- B. comfortable
- C. Luxuries
- D. Indeividual

Answer: B

Solution:

The correct answer is '**comfortable**'.

Key Points

- **Comfortable** is spelt correctly. It means (especially of clothes or furnishings) providing physical ease and relaxation.
 - The correct spellings (and the meanings) of the other words are:
 - **Peirce**--Pierce: to make a hole in something with a sharp point. (नुकीले औज़ार से किसी वस्तु में छेद करना)
 - **Example**-*The needle pierced into her skin.*
 - **Luxuries**--Luxuries: indulgence in and enjoyment of rich, comfortable, and sumptuous living. (विलासिता)
 - **Example**-*The life of the ruling class was comfortable and even luxurious from early times.*
 - **Indeividual**--Individual: single/ separate. (समूह से अलग; अकेला)
 - **Example**-*These styles can be adapted to suit individual tastes.*
 - Hence, the correct answer is **option 2**.
-

Question 56

Select the most appropriate synonym of the given word.

STERN

Options:

- A. strict
- B. mild
- C. lenient
- D. gentle

Answer: A

Solution:

The correct answer is 'strict'.



Key Points

- The most appropriate synonym of the given word 'Stern' is **Strict**.
- **Stern**: very serious; not smiling. (कठोर, बहुत गंभीर; मुस्कान-रहित)
 - *Example: His voice was **stern** again.*
- **Strict**: not allowing people to break rules or behave badly. [सख्त, कठोर (नियम भंग या ग़लत व्यवहार को सहन न करने वाला)]
 - *Example: Samir's very **strict** with his children.*

Correct Answer: strict.



Additional Information

- Let's look at the **meaning** of other words:
 - **mild**: not strong; not very bad. (नरम; हलका)
 - **lenient**: (used about punishment or person who punishes) not as strict as expected. [(दंड या दंड देने वाला व्यक्ति) जो आशा के अनुरूप कठोर न हो; नरम, सौम्य]
 - **gentle**: (used about people) kind and calm; touching or treating people or things in a careful way so that they are not hurt. [(व्यक्ति) दयालु और शांत; बिना चोट पहुँचाए कोमलता से व्यवहार करने वाला; सौम्य]

Question 57

Direction: In the following questions, choose the word opposite in meaning to the given word.

Condemn

Options:

- A. Castigate
- B. Strut
- C. Command
- D. Commend

Answer: D

Solution:

The correct answer is 'Commend'.



Key Points

- The word 'Condemn' is a verb and it means **to criticize something or someone strongly, usually for moral reasons.**
 - For example, *The 26/11 terrorist attack in Mumbai was **condemned** as an act of barbarism and cowardice.*
- The marked option 'Commend' means **to formally praise someone or something.**
 - For example, *The judge **commended** the girl for her bravery.*
- So, we can say that '**Commend**' can function as the opposite word of '*Condemn*.'

Hence, the correct answer is **Option 4**.



Additional Information

- **Castigate** ⇒ to criticize someone or something severely.
- **Strut** ⇒ to walk in a proud, confident gait.
- **Command** ⇒ to order someone (esp in military) or control and responsibility over something.

Question 58

Direction: Choose the most suitable answer to fill the blank:

If you had told me you needed a ride, I _____ earlier.

Options:

- A. would have left
- B. would have been left
- C. will have left

D. will be leaving

Answer: A

Solution:

The correct answer is 'would have left.'

Key Points

- The given sentence is the **type 3 conditional**.
- The **type 3 conditional** is used to refer to a time that is in the past, and a situation that is contrary to reality.
- The structure of **type 3 conditional** is:

If clause	Main clause
If + past perfect	perfect conditional or perfect continuous conditional

- **Example:** *If it had rained you would have gotten wet.*
- Hence, the correct answer is 'would have left.'
- **Correct sentence:** *If you had told me you needed a ride, I would have left earlier.*

Additional Information

- The **zero conditional** is used for when the time being referred to is now or always and the situation is real and possible.
- The **zero conditional** is often used to refer to general truths.
- The tense in both parts of the sentence is the simple present.
- In zero conditional sentences, the word "if" can usually be replaced by the word "when" without changing the meaning.

Hinglish

- दिया गया वाक्य **type 3 conditional** है।
- **Type 3 conditional** का उपयोग उस समय को संदर्भित करने के लिए किया जाता है जो अतीत में है, और ऐसी स्थिति जो वास्तविकता के विपरीत है।
- **Type 3 conditional** की structure है:

If clause	Main clause
If + past perfect	perfect conditional or perfect continuous conditional

- **Example:** *If it had rained you would have gotten wet.*

Question 59

Direction: Choose the most effective word for the blank from the options given below.

Hardly had I reached my office _____ I got your message.

Options:

- A. than
- B. before
- C. after
- D. when

Answer: D

Solution:

The correct answer is '**when.**'



Key Points

- If the second event occurs immediately after the first, we can express that idea using the structure **hardly ... when.**
 - यदि पहली के तुरंत बाद दूसरी घटना होती है, तो हम उस विचार को संरचना **hardly ... when** का उपयोग करके व्यक्त कर सकते हैं।
 - **Example:** *Hardly had I reached the station **when** the train came.*
- **Correct sentence:** *Hardly had I my office **when** I got your message.*



Additional Information

- When we begin a sentence with a negative word, we put the auxiliary verb before the subject.
 - **Example:** *No sooner had she read the letter than she started crying.*
~~*No sooner she read the letter than she started crying.*~~

Question 60

Choose the option with the appropriate question tag for the given sentence:

George has never been to India, _____

Options:

- A. has he?
- B. has George?
- C. hasn't he?

D. hasn't George?

Answer: A

Solution:

The correct answer is 'has he?'.

Key Points

- A **question tag** means a short phrase such as 'isn't it?' or 'did you?' at the end of a sentence that changes it into a question and is often used to ask somebody to agree with you.
 - For example: *She said half price, **didn't she?***
- If the **main clause** is **positive**, the **question tag** is usually **negative**, and if the **main clause** is **negative**, the **question tag** is **positive**.
- The sentence and the question tag must be in the **same tense**.
- For the **negative question tag**, use the **Contracted form of 'helping verb'** and **'not'**
- By referring to the above explanation, it can be concluded that **has he?** should be the appropriate option to be chosen for filling the blank.

Hence, the correct answer is **option 1**.

Question 61

Direction: Choose the option that best transforms the sentence into its other voice:

We added up the money and found it was correct.

Options:

- A. Adding up the money it was found it to be correct by us.
- B. Adding up the money we found it to be correct.
- C. The money was added up and found to be correct.
- D. The money is added up by us and found to be correct.

Answer: C

Solution:

The correct answer is 'The money was added up and found to be correct.'

Key Points

- The given sentence is in **Active Voice**. As per the given question we have to change it into **Passive Voice**.
- The structure of the given sentence is as follows:
 - **Subject + V2 + Object. (Active Voice)**
 - **Subject (objective case) + was/were + V3 + Object (subjective case). (Passive Voice)**
- **Example:**
 - *Their cousin showed them some rare photographs. (Active Voice)*
 - *They were shown some rare photographs by their cousin. (Passive Voice)*
- In the first part of the sentence:
 - **The subject** of the given sentence is '**We**'.
 - **The object** of the given sentence is '**The money**'.
 - **The subject** will be put in place of **the object** and **the object** will be put in place of **the subject**.
 - And '**added up**' will be changed into '**was added up**'.
 - "**Money**" must use "**was**" with it.
- As the second part is connected with the help of the conjunction "**and**" so the same rules will be applied to the verb "**found**."
- Therefore, as per the points mentioned above, we find that the correct answer is **Option 3**.

Correct sentence: *The money was added up and found to be correct.*

Question 62

Direction: Choose the option that best transforms the sentence into its direct / indirect form:

Can the children get tasty meals at school?

Options:

- A. I asked if the children can be getting tasty meals at school.
- B. I asked if the children could get tasty meals at school?
- C. I asked if the children can they get tasty meals at school?
- D. I asked if the children could get tasty meals at school.

Answer: D

Solution:

The correct answer is '**I asked if the children could get tasty meals at school.**'

Key Points

- The given interrogative sentence is in the **direct speech**.
- The rules of **Direct and Indirect speech** for **interrogative sentence** conversions.
 - Inverted commas are removed in indirect speech.

- The reporting verb said/said to is changed to "**asked, demanded, ordered, enquired**" as per the nature of the sentence.
- The first person in the reported speech changes according to the subject of reporting speech.
- In Reported Speech change of the second person depends on the object of reporting speech.
- The third person remains unchanged.
- To make the indirect speech for questions that can be answered in YES/NO, the word "**if**" or "**whether**" is used before the reported speech within the sentence.
- "**Can**" changes to "**could**."
- **Structure:**
 - (Can + Subject+ V1 + Object?) (Direct Speech)
 - (If + Subject + could + V1 + Object) (Indirect Speech)
- **Example:**
 - He said to me "Can you help me?"
 - He asked me if I could help him.
- Therefore, the correct answer is **Option 4**.

Correct sentence: *I asked if the children could get tasty meals at school.*

Confusion Points

- **Option 2** and **4** are different.
- There is a difference in punctuation marks.
- One option has a question mark and the other has a full stop.

Question 63

Arrange the parts(1), (2), (3) and (4) of the following sentences to create meaningful sentences and choose the right option accordingly.

1) people think they're

2) Blackberry phones used to be

3) old-fashioned gadgets

4) extremely popular, but now many

Options:

- A. 4, 3, 1, 2
- B. 3, 1, 2, 4
- C. 4, 1, 3, 2
- D. 2, 4, 1, 3

Answer: D

Solution:

The correct answer is '2, 4, 1, 3'.

Key Points

- The part '2' is independent of any other sentences as it is giving information about the **noun** 'Blackberry phones' that earlier they used to be very popular but with the passage of time, people are considering them old fashioned gadgets. Hence, '2' is the **first part**.
- The **phrase** 'extremely popular' mentioned in the part '4' refers back to the **noun** 'Blackberry phones' mentioned in the part '2'. Hence, '4' follows '2'.
- The **noun** 'people' mentioned in the part '1' is linked with the **phrase** 'but now many' mentioned in the part '4'. Hence, '1' follows '4'.
- The sentence '3' is the **concluding sentence**. Hence, '3' is the **last sentence**.
- The correct sequence is **2, 4, 1, 3**.

Paragraph after rearranging the sentences: -Blackberry phones used to be extremely popular; but now many people think they're old-fashioned gadgets.

Hence, the correct answer is **option 4**.

Question 64

Given here are four jumbled sentence. Pick the option that gives the correct order.

- A.** One day, Maytag's company will create a product
- B.** which will wash 660 million loads of laundry per week
- C.** and become an empire worth 1.7 billion dollars,
- D.** found in over 80 per cent of American households.

Options:

- A. BADC
- B. BDCA
- C. ABDC
- D. ABCD

Answer: D

Solution:

The correct answer is "ABCD"

Key Points

- The first sentence will be A because it introduces Maytag's company and its product
 - **One day, Maytag's company will create a product**
- The second sentence will be B because it states further about the utility of the product
 - **which will wash 660 million loads of laundry per week**
- The third sentence will be C because it states about the empire created by the product
 - **and become an empire worth 1.7 billion dollars.**
- The last sentence will be D because it describes the empire as so huge that the product will be found in every house.
 - **found in over 80 per cent of American households.**

Hence, the correct passage is: *One day, Maytag's company will create a product which will wash 660 million loads of laundry per week and become an empire worth 1.7 billion dollars, found in over 80 per cent of American households.*

Question 65

Select the noun form of the given word:

Different

Options:

- A. Difference
- B. Differ
- C. Differently
- D. Differentness

Answer: A

Solution:

The correct answer is "Difference"

Key Points

- The correct answer is "**Difference**"
- The given word "**Different**" is an adjective that means "**not the same as another or each other; unlike in nature, form, or quality.**"
- "**Difference**" is a noun that means "**the way in which two or more things which you are comparing are not the same**"
- Let's see examples-
 - She seems to wear something **different** every day.
 - What's the **difference** between an ape and a monkey?

Hence, the correct answer is "Option 1 i.e. Difference"

Question 66

Direction: Find out which part has an error and mark it as your answer. If there is no error, mark 'No error' as your answer.

No sooner did she see her teacher (A) / when she (B) / stopped copying. (C) /
No error (D)

Options:

A. (A)

B. (B)

C. (C)

D. (D)

Answer: B

Solution:

The correct answer is '(B).'



Key Points

- The error lies in **Part B** of the sentence.
- In a sentence, the use of **'when'** is incorrect.
- The conjunction **'No sooner ---- than'** is used to denote simultaneous actions.
- **'No sooner'** is always followed by **'than'** and not **'when.'**
 - **Example:** *No sooner had I received her call **than** I left for her place.*
- Therefore, the use of **'when'** in **Part B** of the sentence should be replaced with **'than.'**
- Hence, the correct answer is **'(B).'**

The Correct Sentence: *No sooner did she see her teacher **than** she stopped copying.*



Additional Information

- A **conjunction** is a joiner, a word that connects (conjoins) parts of a sentence.
- **Coordinating conjunctions** join equals to one another.
 - **Example:** *Most children like sweets **and** cookies.*
- Coordinating conjunctions go in between items joined, not at the beginning or end.
 - **Example:** *I like coffee, **but** I don't like tea.*



- दिए गए वाक्यका हिंदी अनुवाद:
 - जैसे ही उसने अपने शिक्षक को देखा उसने नकल करना बंद कर दिया।
 - एक वाक्य में 'when' का प्रयोग गलत है।
 - साथ-साथ होने वाली क्रियाओं को दर्शाने के लिए conjunction 'No sooner ---- than' का प्रयोग किया जाता है।
 - 'No sooner' के बाद हमेशा 'than' आता है न कि 'when'।
-

Question 67

Directions: Read the following passage and answer the given questions:

Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, many argue that these two concepts can coexist and even complement each other. Religion can provide a moral framework and a sense of purpose, while science can explain the natural world and how it operates. One way in which religion and science can coexist is through the concept of intellect. Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it operates. One example of this coexistence is in the study of the origins of the universe. Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other. In conclusion, religion and science are not mutually exclusive. They can coexist and even complement one another, providing a deeper understanding of the world and our place in it. Both offer unique perspectives and insights, and neither necessarily contradicts the other. The key is to approach each with an open mind and a willingness to consider different perspectives.

What is the relationship between religion and science?

Options:

- A. Religion and science have had a simple relationship throughout history.
- B. Science is a systematic method for understanding the natural world.
- C. Religion and science can coexist and even complement each other.
- D. Science is seen as a threat to traditional religious beliefs.

Answer: C

Solution:

The correct answer is "**Religion and science can coexist and even complement each other.**"



Key Points

- Option '3' is correct.
 - From the passage: '*.... Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, many argue that these two concepts can coexist and even complement each other.....*'
 - The passage states that **religion and science can coexist and even complement each other. Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it**

operates.

- Thus, in this context, we can say that the correct answer is option "3"
- Options 1, 2, and 4 are incorrect because there isn't sufficient supportive information in the passage to consider them to be the right answers.

Hence, the correct answer is "Religion and science can coexist and even complement each other."

Question 68

Directions: Read the following passage and answer the given questions:

Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, many argue that these two concepts can coexist and even complement each other. Religion can provide a moral framework and a sense of purpose, while science can explain the natural world and how it operates. One way in which religion and science can coexist is through the concept of intellect. Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it operates. One example of this coexistence is in the study of the origins of the universe. Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other. In conclusion, religion and science are not mutually exclusive. They can coexist and even complement one another, providing a deeper understanding of the world and our place in it. Both offer unique perspectives and insights, and neither necessarily contradicts the other. The key is to approach each with an open mind and a willingness to consider different perspectives.

What is the key to approaching religion and science according to the passage?

Options:

- A. Religion and science can both provide valuable insights and understanding.
- B. The key is to approach each with an open mind and a willingness to consider different perspectives.
- C. The key is to avoid conflating the two; religion and science are mutually exclusive.
- D. Many religious beliefs are not in opposition to scientific facts and theories.

Answer: B

Solution:

The correct answer is "The key is to approach each with an open mind and a willingness to consider different perspectives."



Key Points

- Option '2' is correct.

- From the passage: '.... Both offer unique perspectives and insights, and neither necessarily contradicts the other. **The key is to approach each with an open mind and a willingness to consider different perspectives.....**'
- According to the passage, the key to approaching religion and science is recognizing that they offer different perspectives on the same question and can coexist by considering them as valid ways of understanding and making sense of the world.
- Thus, in this context, we can say that the correct answer is option "2"
- Options 1, 3, and 4 are incorrect because there isn't sufficient supportive information in the passage to consider them to be the right answers.

Hence, the correct answer is "The key is to approach each with an open mind and a willingness to consider different perspectives."

Question 69

Directions: Read the following passage and answer the given questions:

Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, many argue that these two concepts can coexist and even complement each other. Religion can provide a moral framework and a sense of purpose, while science can explain the natural world and how it operates. One way in which religion and science can coexist is through the concept of intellect. Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it operates. One example of this coexistence is in the study of the origins of the universe. Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other.

In conclusion, religion and science are not mutually exclusive. They can coexist and even complement one another, providing a deeper understanding of the world and our place in it. Both offer unique perspectives and insights, and neither necessarily contradicts the other. The key is to approach each with an open mind and a willingness to consider different perspectives.

Select the most appropriate ANTONYM of the given word.
"INTELLECT"

Options:

- A. Genius
- B. Wizard
- C. Stupidity
- D. Sage

Answer: C

Solution:

The correct answer is “Stupidity.”

Key Points

- **"Intellect"** refers to the ability to think and understand things, especially complex or abstract ideas. It can also refer to a person's mental capacity or intelligence. ("बुद्धि" चीजों को सोचने और समझने की क्षमता को संदर्भित करती है, विशेष रूप से जटिल या अमूर्त विचारों को। यह किसी व्यक्ति की मानसिक क्षमता या बुद्धि का भी उल्लेख कर सकता है।)
 - **Example:** "She has a keen *intellect* and was able to understand the complex theory quickly."

Let's see the meanings of the given words

- **"Genius"** refers to a person who is exceptionally intelligent or talented, often used to describe someone who is a prodigy or has exceptional abilities in a specific field. ("जीनियस" एक ऐसे व्यक्ति को संदर्भित करता है जो असाधारण रूप से बुद्धिमान या प्रतिभाशाली है, अक्सर किसी ऐसे व्यक्ति का वर्णन करने के लिए उपयोग किया जाता है जो एक विलक्षण है या किसी विशिष्ट क्षेत्र में असाधारण क्षमता रखता है।)
 - **Example:** "Albert Einstein is considered a *genius* in the field of physics."
- **"Wizard"** refers to a person who is skilled or expert in a particular field, often used to describe someone who is highly proficient in the use of a particular technology or software. It can also refer to a person who has magical powers. ("जादूगर" एक ऐसे व्यक्ति को संदर्भित करता है जो किसी विशेष क्षेत्र में कुशल या विशेषज्ञ है, अक्सर किसी ऐसे व्यक्ति का वर्णन करने के लिए प्रयोग किया जाता है जो किसी विशेष तकनीक या सॉफ्टवेयर के उपयोग में अत्यधिक कुशल होता है। यह किसी ऐसे व्यक्ति को भी संदर्भित कर सकता है जिसके पास जादुई शक्तियां हों।)
 - **Example:** "He is a *wizard* at programming and can fix any problem within minutes."
- **"Stupidity"** refers to the state or quality of being stupid, which means lacking intelligence or common sense. ("मूर्खता" मूर्ख होने की स्थिति या गुणवत्ता को संदर्भित करता है, जिसका अर्थ है बुद्धि या सामान्य ज्ञान की कमी।)
 - **Example:** "The *stupidity* of his decision led to the failure of the project."
- **"Sage"** refers to a person who is wise and has a lot of knowledge and experience, often used to describe someone who is respected for their wisdom and insight. ("ऋषि" एक ऐसे व्यक्ति को संदर्भित करता है जो बुद्धिमान है और जिसके पास बहुत ज्ञान और अनुभव है, अक्सर किसी ऐसे व्यक्ति का वर्णन करने के लिए प्रयोग किया जाता है जो उनकी बुद्धि और अंतर्दृष्टि के लिए सम्मानित होता है।)
 - **Example:** "The old *sage* had lived a long life and had much wisdom to impart to the young people."

According to the meanings of the given words, the correct answer is **Option 3, i.e., Stupidity.**

Question 70

Directions: Read the following passage and answer the given questions:

Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, many argue that these two concepts can coexist and even complement each other. Religion can provide a moral framework and a sense of purpose, while science can explain the natural world and how it operates. One way in which religion and science can coexist is through the concept of intellect. Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it operates. One example of this coexistence is in the study of the origins of the universe. Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other. In conclusion, religion and science are not mutually exclusive. They can coexist and even complement one another, providing a deeper understanding of the world and our place in it. Both offer unique perspectives and insights, and neither necessarily contradicts the other. The key is to approach each with an open mind and a willingness to consider different perspectives.

Give an example of how religion and science coexist in the study of the origins of the universe.

Options:

- A. Many religions have creation myths, while science has the Big Bang theory.
- B. Scientists have used telescopes and other observational techniques to study the history of the universe, while theologians have used scriptures.
- C. The universe is believed to have been created by the god Brahma.
- D. The big bang theory is a scientific explanation for the origins of the universe.

Answer: A

Solution:

The correct answer is "**Many religions have creation myths, while science has the Big Bang theory.**"

Key Points

- Option '1' is correct.
 - From the passage: '*... **Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory.** Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other.....*'
 - The passage states that many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other.
 - Thus, in this context, we can say that the correct answer is option "1"
- Options 2, 3, and 4 are incorrect because there isn't sufficient supportive information in the passage to consider them to be the right answers.

Hence, the correct answer is "*Many religions have creation myths, while science has the Big Bang theory.*"
